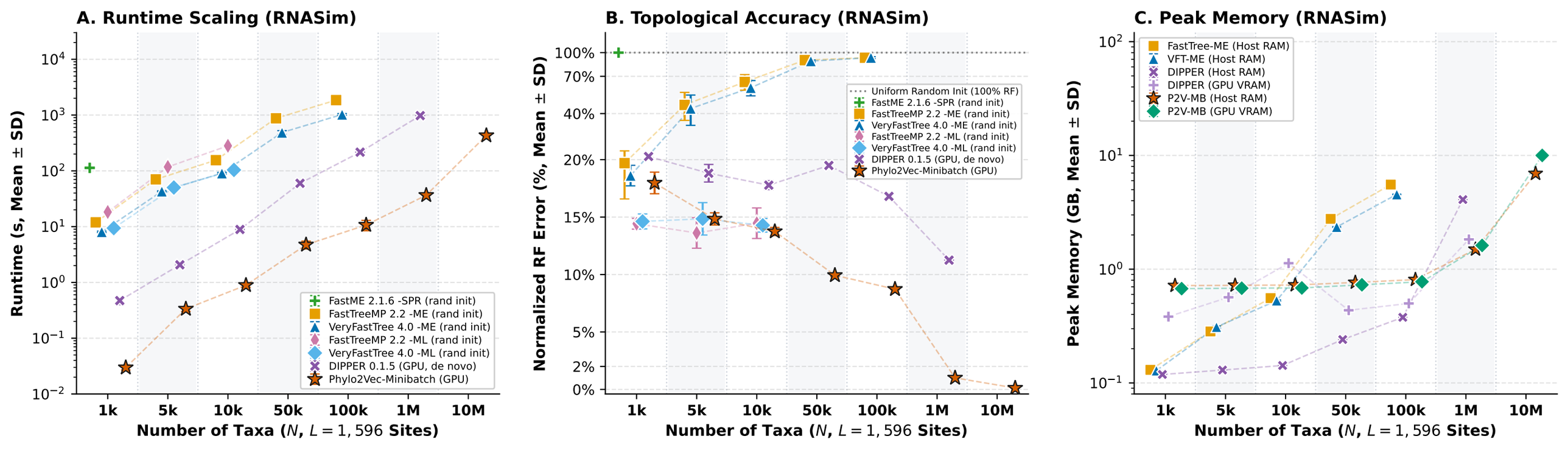
A. Runtime Scaling (RNASim)
Runtime (s, Mean ± SD)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
DIPPER 0.1.5 (GPU, de novo)
Phylo2Vec-Minibatch (GPU)
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 1,596$ Sites)
B. Topological Accuracy (RNASim)
Normalized RF Error (%, Mean ± SD)
100%
70%
40%
20%
15%
10%
5%
2%
0%
Uniform Random Init (100% RF)
FastME 2.1.6 -SPR (rand init)
FastTreeMP 2.2 -ME (rand init)
VeryFastTree 4.0 -ME (rand init)
FastTreeMP 2.2 -ML (rand init)
VeryFastTree 4.0 -ML (rand init)
DIPPER 0.1.5 (GPU, de novo)
Phylo2Vec-Minibatch (GPU)
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 1,596$ Sites)
C. Peak Memory (RNASim)
Peak Memory (GB, Mean ± SD)
10^{2}
10^{1}
10^{0}
10^{-1}
FastTree-ME (Host RAM)
VFT-ME (Host RAM)
DIPPER (Host RAM)
DIPPER (GPU VRAM)
P2V-MB (Host RAM)
P2V-MB (GPU VRAM)
1k
5k
10k
50k
100k
1M
10M
Number of Taxa (N, $L = 1,596$ Sites)